COMP 3920 Database Systems 1

MySQL
Users and Roles

Introduction:
The goal of this document is to serve as a walkthrough creating some users and roles into MySQL 8.

We are going to be using a sample database for a Classic Model Store.
For more information: https://github.com/hhorak/mysql-sample-db

We will be creating 8 users and 4 roles as such:

1. Helen and Henry are in **HR**.
2. Ian and Irene are in **Inventory**.
3. Sally and Sam are in **Sales**.
4. Mary and Mark are in **Management**.

Step 1:
Download `classicmodels MySQL.zip` and unzip.

Step 2:
Use the MySQL `root` account to import the data from inside `classicmodels MySQL.sql`.

Confirm that the database was imported. (What is the name of the database? Replace <DATABASE> with the new database name)

```
SHOW TABLES FROM <DATABASE>;
USE <DATABASE>;
SELECT COUNT(*) FROM products;
```

While still logged in as `root`, create user `Helen`.

```
CREATE USER 'Helen'@'%' IDENTIFIED BY 'password';
```

IMPORTANT: Capitalize the first letter of each user's name.
Example: Helen ***not*** helen.

This creates Helen with access from any host IP address.

Let's do a quick experiment to see what we can do with this.
Log out of MySQL as `root`.
Try to log in as Helen.
What is the result?

Can Helen see the new database?
Can Helen query the products table?

```
SHOW TABLES FROM <DATABASE>;
USE <DATABASE>;
SELECT COUNT(*) FROM products;
```

Can Helen change her own password?

```
ALTER USER Helen IDENTIFIED BY 'password2';
```

If yes, change the password back.

Log out of MySQL as `Helen`.

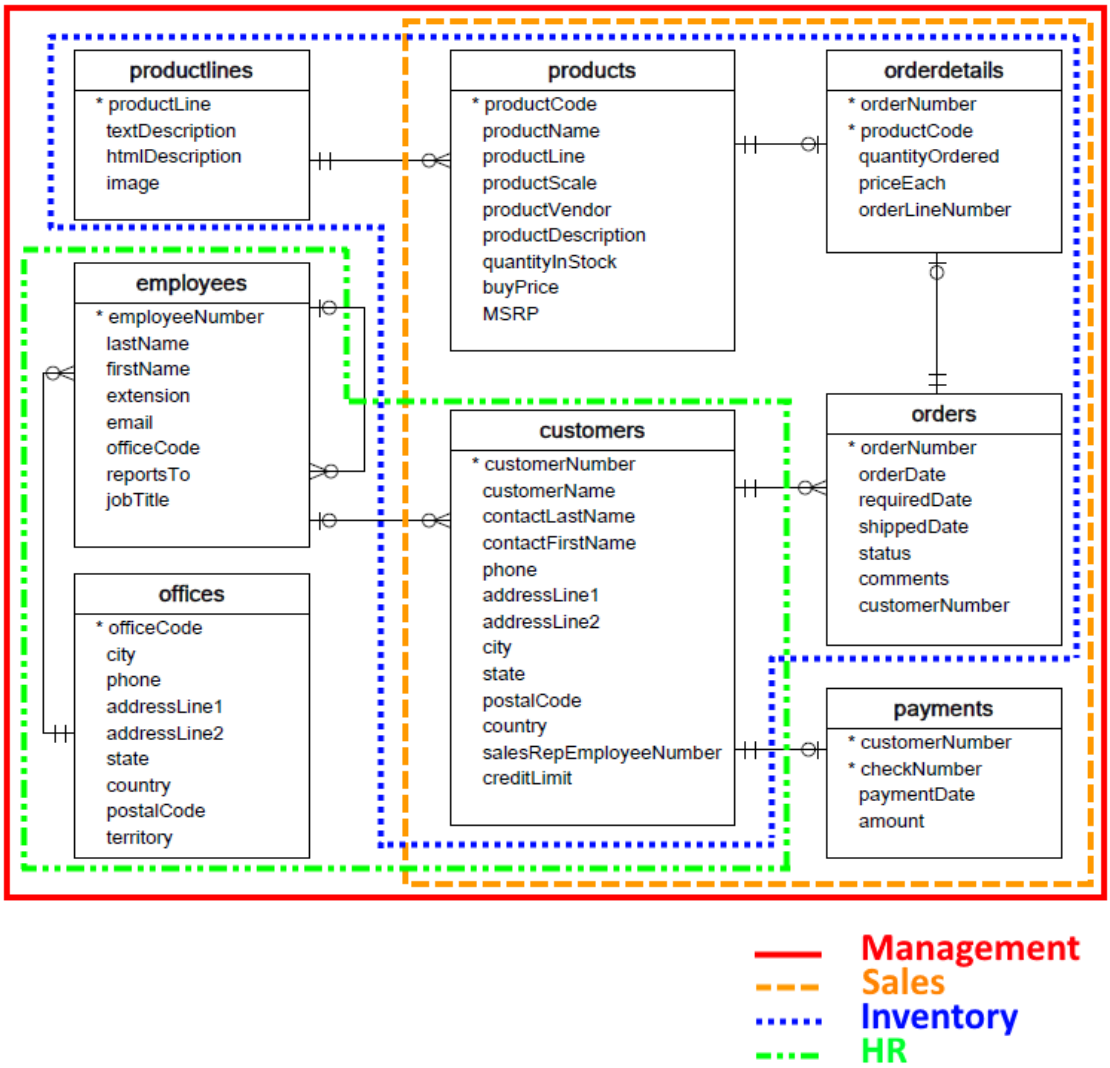
What is your conclusion?

Step 3:
Log back in to MySQL as `root` to finish creating all the other users (8 in total – 7 more).

Step 4:
Create the 4 roles:
HR, Inventory, Sales and Management.
Ex: `CREATE ROLE 'HR';`

IMPORTANT: Capitalize the first letter of each role's name.
Example: Inventory ***not*** inventory.

Step 5:
Grant the following privileges to each role as shown in the diagram below:
Privileges to grant: SELECT, INSERT, UPDATE, DELETE.



Ex: For the `HR` role we need to grant `SELECT`, `INSERT`, `UPDATE` and `DELETE` on the `employees`, `customers` and `offices` tables.

Step 6:
Assign each of our 8 users their appropriate roles:

1. Helen and Henry are in **HR**.
2. Ian and Irene are in **Inventory**.
3. Sally and Sam are in **Sales**.
4. Mary and Mark are in **Management**.

Ex: `GRANT 'HR' TO 'Helen';`

Step 8:
Activate the roles on all 8 users.
Because any role for a particular user MySQL can be active or inactive, we need to activate all roles for all our users so they take effect.
Ex: `SET DEFAULT ROLE ALL TO 'Helen';`

Run the following command:

```
mysql -u root -p -t < "Verify Users and Roles MySQL.sql" >
"01_Results_MySQL.txt"
```

IMPORTANT! Make sure you include the **`-t`** option when running the above command.

Submit your results file:
`01_Results_MySQL.txt`

Submit

The initial contents of `01_Results_MySQL.txt` should look like this:

```
+-------------+---------------+---------------+--------------+----------------+--------------+
| GRANTEE     | TABLE_CATALOG | TABLE_SCHEMA  | TABLE_NAME   | PRIVILEGE_TYPE | IS_GRANTABLE |
+-------------+---------------+---------------+--------------+----------------+--------------+
| 'HR'@'%'    | def           | classicmodels | customers    | DELETE         | NO           |
```